

Shaik Riyaz Basha

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PROFESSIONAL SUMMARY

Final-year B.Tech CS (AI) student specializing in backend & distributed systems, with hands-on experience building scalable microservices and cloud-native applications. AWS Certified Solutions Architect with strong foundations in system design (LLD/HLD), design patterns, and performance optimization.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL

Frontend: React, Next.js, Tailwind CSS, WebSocket

Backend: Node.js, Express.js, REST APIs, Microservices, RabbitMQ, Redis

Databases: MongoDB, PostgreSQL

Cloud & DevOps: AWS (EC2, S3, RDS, IAM, VPC), Docker, Linux

HLD: Distributed Systems, Microservices, Event-Driven Architecture, Caching, Rate Limiting, Load Balancing, API Design

LLD: SOLID Principles, Design Patterns (Singleton, Strategy, Observer, Factory, Builder, State), OOP

Tools: Git, GitHub, Docker, Postman, VS Code, Cursor, AWS Console, Vercel

EDUCATION

G. Pullaiah College of Engineering & Technology

B.Tech in Computer Science and Engineering (AI)

Expected May 2027

CGPA: 8.44 / 10.0

TECHNICAL PROJECTS

Self-Healing Distributed Cache

Node.js, TypeScript, Express.js, Docker

Jun 2026 – Present

[GitHub](#)

- Engineered a fault-tolerant distributed cache across 3 nodes using circular replication topology, ensuring less data loss on node failure via primary/replica key distribution.
- Built a self-healing coordinator that detects failures via heartbeat monitoring, auto-restarts dead nodes using Node.js `child_process`, and restores data from surviving replicas.
- Integrated LRU eviction policy using insertion-order-aware `Map`, evicting least-recently-used entries under memory pressure to prevent unbounded cache growth.
- Reduced per-key memory footprint by ~60% by storing only (**key, value, type**) per entry and offloading ownership resolution to coordinator topology.

Real-Time Chat Application (Microservices)

Node.js, RabbitMQ, Redis, WebSocket, MongoDB, TypeScript

Dec 2025 – Jan 2026

[GitHub](#) | [Live](#)

- Architected distributed microservices with RabbitMQ (including dead-letter queues) ensuring reliable event-driven communication across services, with RabbitMQ containerized via Docker for consistent and portable deployment.
- Engineered WebSocket-based messaging system supporting 500+ concurrent users using Redis caching, rate limiting, and pub/sub messaging patterns.
- Reduced server load by 40% via MongoDB query optimization, collection indexing, and connection pooling strategies.

URL Shortener

Next.js, TypeScript, Node.js, Express.js, PostgreSQL, Redis

Feb 2026 – Mar 2026

[GitHub](#) | [Live](#)

- Implemented URL shortening engine using Redis atomic `INCR` for collision-free ID generation combined with Base62 encoding, replicating the core architecture used by production-grade link shorteners.
- Designed MVC-style backend with Repository pattern using raw PostgreSQL queries, eliminating ORM overhead and improving query performance.
- Secured the API with JWT authentication, bcrypt password hashing, Helmet HTTP headers, and express-rate-limit middleware to prevent abuse at scale.

ACHIEVEMENTS

- **1st Place** in an inter-college coding competition (700+) and **1st Rank** in a college coding assessment (350+).
- Volunteered as **Technical Team Member** at **AI Expo 2026**, supporting technical operations and event coordination.
- **LeetCode** : 250+ problems solved across DSA fundamentals

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (Jun 2026)

AWS Cloud Practitioner Essentials (Dec 2025)